

EMBEDDED HACKING

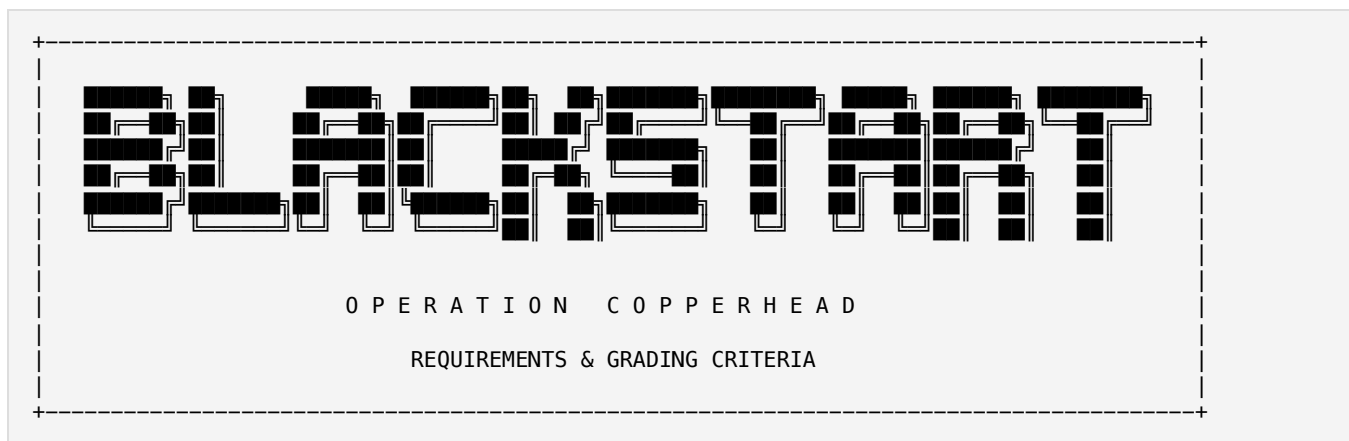
Reverse Engineering the Raspberry Pi Pico 2

Operation Copperhead - Requirements & Grading Criteria

ARM Cortex-M33 • GDB • Ghidra

October 1, 2026

For educational and defensive security research only



Project Overview

Students are the reverse-engineering reserve team called in after DEEPLINE Metro Authority's rebuilt DEEPLINE-AUTH relay image shipped four corrupted engineering constants: a miscalibrated release threshold, a false TRACK banner string, an overstated block length, and a poisoned ARX signal seed. Students reverse engineer `CTF-02.bin` with Ghidra, patch all four defects, capture the runtime-derived signal key live in GDB, recover the Ouroboros authority frame, export a corrected image, flash it to real hardware, and prove the corrected behavior on a physical Pico 2.

The challenge is a standalone capstone exercise and contains no answer, constant, address, bug, or patch belonging to any other course assignment.

Learning Objectives

- Decode an ARM Cortex-M33 vector and boot table and identify the reset handler and initial stack pointer.
- Translate Thumb reset-vector addresses into real function entry points and trace literal-pool entries to their data.
- Locate four corrupted constants: a boundary comparison, a status string, an 8-byte IEEE-754 double, and an ARX signal seed.
- Capture a runtime-derived key with GDB, override a register, and set a watchpoint on stored SRAM state.
- Recover and authenticate an Argon2id plus XChaCha20-Poly1305 authority frame.

Students must use only Weeks 1-8 concepts: ARM registers, stack behavior, USB-CDC output, GDB, Ghidra static analysis and binary patching, vector tables, reset startup, XIP, Thumb addressing, data segments and literal pools, condition-code analysis, runtime key derivation, and the Argon2id plus XChaCha20-Poly1305 authenticated gate.

Deliverables Checklist

#	Deliverable	Format	Criterion
1	Ghidra project screenshot	PNG/JPG	1.1
2	Vector table and boot table	Inside <code>CTF-02-Answers.md</code>	1.2

#	Deliverable	Format	Criterion
3	<code>main()</code> and status-loop table	Inside <code>CTF-02-Answers.md</code>	1.3
4	Literal pool trace	Inside <code>CTF-02-Answers.md</code>	1.4
5	Bug #1 evidence and patches	Inside <code>CTF-02-Answers.md</code>	2.1-2.4
6	Bug #2 evidence and patch	Inside <code>CTF-02-Answers.md</code>	3.1-3.4
7	Bug #3 evidence and patch	Inside <code>CTF-02-Answers.md</code>	4.1-4.3
8	Bug #4 evidence and patch	Inside <code>CTF-02-Answers.md</code>	5.1-5.4
9	GDB register capture	Inside <code>CTF-02-Answers.md</code>	6.1-6.4
10	Ouroboros gate recovery and auth	Inside <code>CTF-02-Answers.md</code>	7.1-7.4
11	<code>CTF-02_fixed.bin</code>	BIN file	8.1
12	<code>CTF-02_fixed.uf2</code>	UF2 file	8.2
13	Corrected console transcript	Inside <code>CTF-02-Answers.md</code>	8.3
14	Summary table of all patches	Inside <code>CTF-02-Answers.md</code>	8.4
15	Written reflection	Inside <code>CTF-02-Answers.md</code>	9.1-9.2

Required Tools and Equipment

Tool	Purpose
Raspberry Pi Pico 2	Isolated target
USB-CDC virtual serial console	Observe output and type the gate passphrase
SWD debug probe	GDB inspection
Ghidra	Static analysis and binary patching
GDB	Dynamic analysis and register capture
Python (<code>uf2conv.py</code>)	UF2 conversion
<code>CTF-02.bin</code> and <code>CTF-02.uf2</code>	Supplied artifacts

Console settings: **USB-CDC virtual COM port, 115200 baud, 8 data bits, no parity, 1 stop bit.**

Artifact Identity

The instructor-issued artifact hashes are:

```
CTF-02.bin  85330C37CD0897746B1AF447E4BAC371DDE2042ABD2D61D58A61FE2A8EEF3537
CTF-02.uf2  F3CD4840260DB820D792758CECACC5297BEF1971B9EACF7601279256D8AF1EAB
```

Grading Rubric - Detailed Breakdown

Task 1: Setup and Initial Analysis (12 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 1.1: Ghidra Project Setup	3	Correct project name, ARM:LE:32:Cortex (ARM Cortex 32-bit little endian), base 0x10000000	One item off	Not set up
Criterion 1.2: Vector Table Decoding	3	Correct base, initial SP, reset pointer	One missing	Not found
Criterion 1.3: main() and Status-Loop Addresses	4	Both addresses correct	One correct	Neither found
Criterion 1.4: Thumb Addressing and Literal Pool	2	Bit 0 cleared and one pool entry traced to its string	Partial	Incorrect

Task 2: Find and Patch Bug #1: The Miscalibrated Release Threshold (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 2.1: Locate Compare Sites A and B	6	Both addresses and original bytes	One site	Not found
Criterion 2.2: Correct Immediate-Value Reasoning	4	Explains the < to <= transform and gives 0x3B	Correct value, no reasoning	Wrong value
Criterion 2.3: Patch Compare Sites A and B	4	Both byte changes verified	One site	Not patched
Criterion 2.4: Explain Why Both Sites Must Be Patched	1	Clear explanation of the two independent comparisons	Vague	Missing

Task 3: Find and Patch Bug #2: The False TRACK Banner (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 3.1: Locate the Banner String	3	Correct address and cross-reference	Approximate	Not found
Criterion 3.2: Patch Six Characters	4	All six bytes changed, length preserved	Correct text, wrong bytes documented	Wrong length
Criterion 3.3: Character-by-Character Documentation	2	Original vs patched byte for all six characters	Partial	Missing
Criterion 3.4: Explain the Danger of a Hardcoded Status Word	1	Clear, specific reasoning	Generic	Missing

Task 4: Find and Patch Bug #3: The Block Length Constant (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 4.1: Locate the .data Double	3	Correct address traced from the BLOCK LENGTH print	Approximate	Not found
Criterion 4.2: IEEE-754 Bytes and Print Math	4	Original and patched 8-byte double with print math (3200 M to 320 M)	Correct patch, no math	Wrong bytes
Criterion 4.3: Patch to Print 320 M	3	Console shows <code>BLOCK LENGTH: 320 M</code>	Wrong bytes	Not patched

Task 5: Find and Patch Bug #4: The Signal Seed (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 5.1: Locate SIGNAL_SPEC and the Seed	5	<code>0x2D879291</code> located and wrong seed <code>0x0A0A0A0A</code> found	Partial	Not found
Criterion 5.2: Patch the Seed	4	Seed bytes changed to <code>74 65 20 6B</code>	Wrong byte	Not patched
Criterion 5.3: Explain the ARX Derivation	3	Correct trace of the per-cycle derivation	Vague	Missing
Criterion 5.4: Separate the Security Layers	3	Correctly explains what the seed fixes versus the gate	Generic	Missing

Task 6: GDB Register Capture of the Derived Key (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 6.1: Breakpoint at the Derive Return	4	Correct address and <code>\$r0</code> read as the bug-derived key	Address off	Not found
Criterion 6.2: Inspect the Two Arguments	4	Live seed and derived IV captured at the second call	One correct	Missing
Criterion 6.3: Override the Register	4	<code>\$r0</code> set to <code>0x2D879291</code> and the next cycle shows <code>OK</code>	Partial	Missing
Criterion 6.4: Watchpoint on the Stored Key	3	Watchpoint on the SRAM key location documented	Approximate	Missing

Task 7: Recover the Ouroboros Authority Frame (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 7.1: Locate Salt, Nonce, Ciphertext, and Tag	4	All three addresses correct in flash	Two correct	Not found

Criterion	Points	Full credit	Partial credit	No credit
Criterion 7.2: Document Argon2id Parameters and Payload Contract	2	Correct memory/time/parallelism and payload layout	Partial	Missing
Criterion 7.3: Authenticate with the 12-Word Passphrase	2	AUTHORITY FRAME: VERIFIED , LED on, payload printed	Partial	Not shown
Criterion 7.4: State the Honest Quantum Boundary	2	Grover halves symmetric exponents; not strict PQC	Generic	Misstates

Task 8: Export and Verify (8 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 8.1: Export CTF-02_fixed.bin	1	Valid patched binary	Corrupted	Not submitted
Criterion 8.2: Convert to CTF-02_fixed.uf2	1	Correct base and family flags	Wrong flags	Not submitted
Criterion 8.3: Hardware Verification	4	Corrected console output confirmed (CRITICAL/HELD in 2s stream; DANGER at boot/Ghidra)	Some lines corrected	No verification
Criterion 8.4: Summary Table of All Patches	2	Complete address and before/after table	Missing entries	No table

Task 9: Written Reflection (5 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 9.1: "Rushed Build" Is Not an Excuse	2	Specific, grounded reasoning	Generic	Missing
Criterion 9.2: One Engineering Practice per Failure Area	3	Concrete practices for the bugs and for image authenticity	Names some	Missing

Common Pitfalls

Pitfall	Consequence	Avoidance
Patching only one threshold site	One status line still lies	Patch both <code>0x10000302</code> and <code>0x10000312</code>
Assuming the immediate equals the limit	Off-by-one, wrong boundary	Use <code>0x3B</code> (59), not <code>0x3C</code> (60)
Using Patch Instruction before IT block	Re-disassembler context conflict swallows Site B	In Listing press <code>C</code> -> edit byte in Bytes window (pencil) -> press <code>D</code>

Pitfall	Consequence	Avoidance
Missing boot banner in serial terminal	PuTTY misses one-time 5ms boot banner	Pulse RUN to GND while connected to capture
Replacing a string with a different length	Corrupts adjacent flash	<code>NORMAL</code> and <code>DANGER</code> are both 6 bytes
Treating the block length as an integer	Misses the 8-byte double	Follow the value into <code>.data</code> , decode IEEE-754
Using the wrong 0.32 bytes	Prints 316 M instead of 320 M	Use <code>7B 14 AE 47 E1 7A D4 3F</code>
Treating an odd vector address as invalid	Thumb analysis fails	Clear bit 0
Starting the seed patch at the wrong offset	Wrong seed, key never matches	Seed is at <code>0x1000EC70</code>

How To Breadboard

- **Raspberry Pi Pico 2** powered over USB.
- **USB-CDC virtual serial console:** open the Pico's COM port at 115200 baud, 8 data bits, no parity, 1 stop bit.
- **SWD debug probe:** connect SWCLK, SWDIO, GND, and 3.3 V to the Pico debug header for GDB inspection and register capture.
- No other peripherals are required; the authority LED is on-board.

Memory Map Reference

Region	Address	Purpose
Bootrom	<code>0x00000000</code>	Immutable boot code
Flash/XIP	<code>0x10000000</code>	Vector table, code, rodata, <code>.data</code> init image
SRAM	<code>0x20000000</code>	Stack and writable state

Deadline & Submission

- Create a folder containing the Ghidra screenshot, `CTF-02_fixed.bin`, and `CTF-02_fixed.uf2`.
- Write all written answers in a single file named `CTF-02-Answers.md` inside that folder.
- ZIP the folder as `lastname-firstname-CTF-02.zip`.
- Submit the ZIP before the posted deadline; late submissions lose 10 percent per day.

Grade Scale

Grade	Percentage	Points
A+	97-100%	97-100
A	93-96%	93-96
A-	90-92%	90-92
B+	87-89%	87-89
B	84-86%	84-86
B-	80-83%	80-83
C	70-79%	70-79
F	0-69%	0-69

Academic Integrity

Use only the supplied Pico 2 and firmware. Do not connect the exercise to an operational railway, metro system, public network, military system, or third-party device. This is a controlled, isolated educational exercise. All analysis and patches must be your own work; sharing binaries, addresses, keys, passphrases, or answers is a violation of the academic integrity policy.

Reference Material

Topic	Reference
ARM Cortex-M33 registers and stack	Week 1
USB-CDC output and console capture	Week 2
Vector tables, reset startup, and XIP	Week 2
Ghidra static analysis and binary patching	Week 3
Data segments, literal pools, IEEE-754	Week 4
Condition-code analysis	Week 5
Runtime key derivation and GDB register capture	Week 6
Argon2id and XChaCha20-Poly1305 authenticated gate	Week 8